

Project 1: Architectural Case Study - Sovereign Cloud Infrastructure & Access Governance

 **Portfolio Methodology Note:** This project profile represents a retrospective technical analysis of a real-world sovereign cloud deployment that I supported as an Advisor. While the physical implementation and code deployment were executed by the Core Systems Engineering division, this document demonstrates my technical aptitude to analyze, map, and architect the end-to-end security control logic of the environment through a professional Governance, Risk, and Compliance (GRC) lens.

Executive Case Overview

- **Domain:** Perimeter Security, Identity & Access Governance, Public Sector Cloud Isolation
- **Target Framework Benchmarks:** NIST SP 800-53 Rev. 5, FedRAMP Moderate Baseline
- **Objective:** Technical deconstruction and control-mapping of a high-stakes cloud environment deployment, enabling a global telecommunications carrier to securely deliver authorized solutions to US federal agencies.

The Structural & Regulatory Hurdles

During a multi-entity project deployment, process integration friction between vendor platforms and carrier data routing interfaces threatened project timelines. From a GRC perspective, the environment was bound by strict **sovereign security mandates** requiring:

1. Complete system isolation operating strictly within domestic boundaries to prevent cross-border data exposure.
 2. An ironclad operations framework restricting data visibility exclusively to cleared US citizen personnel.
 3. Unified ticketing systems across disparate corporate networks without leaking protected system metadata.
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GRC Methodology & Security Control Mapping

1. Identity & Access Enforcement (NIST SP 800-53: AC-2, AC-3)

- **The GRC Approach:** To prevent unauthorized or non-cleared personnel from accessing public sector data streams, a GRC professional must enforce a rigid **Role-Based Access Control (RBAC)** schema that intercepts incoming service tickets.
- **The Logic Implementation:** I mapped the explicit business logic requirements for an identity access validation gateway. This design requires the system to programmatically cross-reference incoming user directory tags and automatically flag or block access for any user not verified in the sovereign staff directory.

2. Boundary Protection & Gateway Routing (NIST SP 800-53: SC-7, SA-9)

- **The GRC Approach:** Ensuring data integrity requires continuous validation of telemetry data pathways and network connection nodes.
- **The Logic Implementation:** I evaluated the data path architectures to verify that all payloads originating from federal agency assets are structurally segregated from commercial database clusters and route exclusively through secure, domestic hosting gateways.

3. Compliance Governance & SOP Lifecycle (NIST SP 800-53: CM-9)

- **The GRC Approach:** Eliminating operational risk during multi-asset transitions requires moving from a "people-dependent" model to a standardized "process-dependent" model.
- **The Logic Implementation:** I structured the formal hand-off boundaries linking End-Customer Advisors, Technical Support Engineers, and the Partner Platform Lead to ensure a stabilized, audit-ready steady state.